

The empirical recurrence for OEIS A221993, proved by transfer matrix

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Abstract

OEIS A221993 carries an empirical recurrence of order 9 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is local to a set of automaton states – those the input can have left the marker in, one such state being the last $|K|$ input values together with one bit saying whether the nearest earlier position whose convolved value differs from the current one is lower – sets with the same future behaviour being identified, so the admissible configurations of such a window are the vertices of a finite digraph and the arrays counted are exactly the walks in it; hence $a(n)$ is a walk count on $S = 11$ vertices and is C -finite. Whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A221993 is “Number of binary arrays indicating the locations of trailing edge maxima of a random length- n 0..2 array extended with zeros and convolved with 1,4,6,4,1.”. It has offset 1 and begins

$$2, 3, 4, 6, 10, 15, 22, 36, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

$$\text{Empirical: } a(n) = a(n-1) + 3*a(n-3) - 2*a(n-4) - 3*a(n-6) + 2*a(n-7) + a(n-9).$$

As of the “Last modified” line on the live entry (Aug 14 2018 08:57:58, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The entry counts distinct IMAGES and not the objects that produce them, so the vertices are not single configurations but sets of them: a set of automaton states – those the input can have left the marker in, one such state being the last $|K|$ input values together with one bit saying whether the nearest earlier position whose convolved value differs from the current one is lower – sets with the same future behaviour being identified. The set a word leaves the model in is determined by the word, and an edge from u to v carries the symbol emitted in passing, so distinct output words and walks from the initial vertex are the same thing. The condition itself is decided within the model by inspection of a state and the symbol read. A distinct indicator array is then exactly a walk of length n followed by one of the distinct completions forced by the

all-zero tail beyond the input; ι is the indicator of the initial vertex and τ is the vector counting, at each vertex, those completions, so with M the adjacency matrix on $S = 11$ vertices,

$$a(n) = \iota^\top M^n \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C -finite.

3 The proof

Lemma 1. *Let $q(t) = t^9 - \sum_i c_i t^{9-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+9} - \sum_i c_i M^{j+9-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = a(n-1) + 3a(n-3) - 2a(n-4) - 3a(n-6) + 2a(n-7) + a(n-9)$$

for every $n > 0$.

Proof. The vector $w = q(M)\tau$ was formed by 9 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 0 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

4 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 39 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
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